

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Ratios, rates, proportional relationships, and units

01

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

A group of monarch butterflies migrated from Chicago, Illinois, to Michoacán, Mexico, flying a total of 2,100 miles. It took a single butterfly in the group 120 days to travel this route one way. On average, how many miles did the butterfly travel per day?

- A. 0.057
- B. 0.729
- C. 17.5
- D. 24

Q2

PROBLEM-SOLVING AND DATA ANALYSIS

To make a bakery's signature chocolate muffins, a baker needs 2.5 ounces of chocolate for each muffin. How many pounds of chocolate are needed to make 48 signature chocolate muffins? (1 pound = 16 ounces)

- A. 7.5
- B. 10
- C. 50.5
- D. 120

Q3

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

How many yards are equivalent to 77 rods? $5.5 \text{ yards} = 1 \text{ rod}$

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

The population density of Worthington is 290 people per square mile. Worthington has a population of 92,800 people. What is the area, in square miles, of Worthington?

- A. 102,400
- B. 93,090
- C. 320
- D. 32

Q5

PROBLEM-SOLVING AND DATA ANALYSIS

The weight of an object on Venus is approximately $\frac{9}{10}$ of its weight on Earth. The weight of an object on Jupiter is approximately $\frac{23}{10}$ of its weight on Earth. If an object weighs 100 pounds on Earth, approximately how many more pounds does it weigh on Jupiter than it weighs on Venus?

- A. 90
- B. 111
- C. 140
- D. 230

What length, in centimeters, is equivalent to a length of 51 meters?

1 meter = 100 centimeters

- A. 0.051
- B. 0.51
- C. 5,100
- D. 51,000

Q7

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A special camera is used for underwater ocean research. When the camera is at a depth of 58 fathoms, what is the camera's depth in feet? 1 fathom = 6 feet

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In a box of pens, the ratio of black pens to red pens is 8 to 1. There are 40 black pens in the box. How many red pens are in the box?

- A. 5
- B. 8
- C. 40
- D. 320

Q9

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A product costs 11.00 dollars per pound. What is the cost, in dollars, for 6 pounds of the product?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Marta has 7,500 pesos she will convert to US dollars using a currency exchange service. At this time, the currency exchange rate is 1 peso = 0.075 US dollars. The exchange service will charge Marta a 2% fee on the converted US dollar amount. How many US dollars will Marta receive from the currency exchange after the 2% fee is applied?

- A. \$551.25
- B. \$562.50
- C. \$5,625.00
- D. \$98,000.00